

Chapter

Constructors

1. A special method that initializes values to data members is called _____.

- (a) Constructor
- (b) Method
- (c) Function
- (d) All of the above **MAIN 2024**

2. Is the following statement true/false? Access specifiers can be used for defining constructors. SQP 2023

- (a) True
- (b) False
- (c) May be
- (d) May not be

3. Which of the following is a return type of a constructor? COMP 2021

- (a) Void
- (b) Int
- (c) Null
- (d) None of the above

4. What is the primary function of a parameterized constructor? MAIN 2021

- (a) To return a value
- (b) To set default values
- (c) To accept arguments for initialization
- (d) To invoke another constructor

5. Which of these is an example of constructor overloading? COMP 2021

- (a) A class having multiple constructors with the same name and parameter lists.

- (b) A class having multiple constructors with different parameter lists.
- (c) A class having a single constructor.
- (d) A class having private constructors only.

6. (i) How many constructors can a class have? **SQP 2024**

(ii) Why do constructors not have any return type?

7. Can the number of actual parameters be different from the number of formal parameters? **SQP 2021**

8. If a programmer does not define a constructor in a class, can't the programmer be able to create an object in the program? **COMP 2022**

9. Create a class with one integer instance variable. Initialize the variable using: (i) Default constructor

(ii) Parameterized constructor **MAIN 2023**

Solution

1. (a) Constructor

A constructor is a special method in Java that is automatically called when an object is created. Its main purpose is to **initialize the data members** of the class.

2. (a) True

Yes, access specifiers like **public, private, and protected** can be used with constructors to control how objects of a class are created.

3. (d) None of the above

A constructor **does not have any return type**—not even void. It only initializes the object when it is created.

4. (c) To accept arguments for initialization

A parameterized constructor allows values to be passed when an object is created so that the data members can be initialized with specific values.

5. (b) A class having multiple constructors with different parameter lists.

Constructor overloading means a class can have more than one constructor, but each must have a different number or type of parameters.

6. (i) A class can have **any number of constructors**, as long as each constructor has a **different set of parameters** (different number or types).

This is called **constructor overloading**.

(ii) Constructors do **not have any return type**, not even void, because their **implicit return type is the object (class type) itself**.

Their purpose is to initialize objects, not to return values.

7. The **number of actual parameters** passed while calling a constructor must **match the number of formal parameters** defined in that constructor. If they do not match, the compiler will produce an **error**, unless a different constructor (with a matching parameter list) is available.

8. If a programmer does not define any constructor in a class, the **compiler automatically provides a default constructor**. This default constructor allows the creation of objects and initializes instance variables to their default values (e.g., 0 for numbers, false for boolean, null for objects).

9.

```
class Test {
    int a;

    Test() {
        // variable initialization
using default constructor
        a = 0;
    }

    Test(int x) {
        // variable initialization
using parameterized constructor
        a = x;
    }
}
```